Hierarchical organization of cognitive processes underlies low cross-task correlations in inhibitory control: Evidence from drift diffusion modeling

HIERARCHICAL PROCESSES IN INHIBITORY CONTROL

Abstract

Tasks measuring inhibitory control consistently yield low cross-task correlations, yet whether this pattern reflects genuine construct heterogeneity or the conflation of distinct cognitive processes within traditional behavioral indices remains unresolved. Here, 125 participants completed three interference control tasks (*Stroop*, *Flanker*, *Simon*) and three response-inhibition tasks (*Go/No-Go*, *Stop-Signal*, *Antisaccade*) across two independent sites. Hierarchical Bayesian drift diffusion modeling decomposed trial-level performance into drift rate (v), decision threshold (a), non-decision time (t), and starting point bias (z); cross-task sharing was examined through confirmatory factor analysis, partial correlations, and machine-learning ablation, with cross-site replication. The four parameters followed distinct cross-task patterns. Drift rate, decision threshold, and non-decision time were broadly shared across both task types, each well characterized by a single-factor model. Starting point bias, in contrast, showed selectively stronger covariation within response inhibition tasks, where the two-factor model significantly outperformed the single-factor model. Condition-difference drift rate (Δv) showed only low-to-moderate cross-task correlations, comparable to traditional interference effects. Critically, controlling for drift rate substantially reduced cross-task correlations among traditional indices, indicating that much of their shared variance reflects domain-general processing efficiency rather than inhibition-specific processes. The drift rate factor also incrementally predicted fluid intelligence beyond traditional indices. These findings indicate that inhibitory control engages hierarchically organized cognitive processes: domain-general efficiency is shared broadly across task types, whereas response tendency modulation is shared selectively within response inhibition. Traditional difference scores collapse these distinct levels into single values, diluting shared signals. Low cross-task correlations thus largely reflect measurement masking rather than genuine construct fragmentation.

Keywords: inhibitory control; drift diffusion model; individual differences; evidence accumulation; cross-task consistency

Public Significance Statement

Inhibitory control—the ability to override unwanted impulses—is widely used to assess conditions such as ADHD, addiction, and cognitive aging, yet standard laboratory tasks designed to measure it often fail to agree with one another. This study shows that the disagreement arises not because the ability is illusory, but because conventional scoring methods blend together separable mental processes that follow different organizational patterns across tasks.

Introduction

Inhibitory control is among the most pervasive constructs in psychology. From predicting children's academic achievement to assessing cognitive deficits in psychiatric disorders, from explaining individual differences in self-regulation to tracking cognitive decline in aging, this construct has been invoked across domains spanning cognitive science, developmental psychology, and clinical assessment (Diamond, 2013; Duckworth & Seligman, 2005; Miyake et al., 2000; Moffitt et al., 2011; Snyder et al., 2015). Yet the empirical foundation supporting it harbors a well-documented puzzle: tasks designed to measure inhibitory control show remarkably low correlations with one another. This pattern is not an anomalous finding but a systematic empirical regularity. Rey-Mermet et al. (2018) examined 11 inhibitory control tasks and found only weak inter-task correlations coupled with poor structural equation model fit. Alternative scoring approaches—including accuracy indices, inverse efficiency scores, and bifactor models—have all failed to extract stable inhibition-specific variance (Gärtner & Strobel, 2021; Rey-Mermet et al., 2020). Within the Miyake framework, variance in the inhibition factor is completely absorbed by a common executive function factor, leaving no separable inhibition component (Friedman et al., 2008; Miyake & Friedman, 2012). Although recent large-sample studies have managed to re-isolate a response inhibition factor at the behavioral level, they continue to rely on traditional difference scores (Shields & Yonelinas, 2025). Importantly, this pattern persists even after correcting for measurement unreliability, indicating that poor reliability alone cannot fully account for the low cross-task correlations (Hedge et al., 2018; Rey-Mermet et al., 2018).

This pattern presents researchers with a fundamental theoretical dilemma. One possibility—the construct failure hypothesis—holds that inhibitory control is a mistakenly unified label, with "interference control" and "response inhibition" indexing qualitatively different cognitive operations that share no common foundation. The alternative—the measurement masking hypothesis—holds that cognitive processes shared across tasks do exist, but traditional indices conflate multiple processes with different sharing scopes into a single value, thereby obscuring the common components that are in fact present (Draheim et al., 2019). Traditional behavioral indices alone cannot adjudicate between these possibilities—the single value produced by a subtraction operation cannot reveal which cognitive processes are conflated within it, nor what cross-task organizational patterns each process follows (Cronbach & Furby, 1970; Miller & Ulrich, 2013). Resolving this dilemma requires an analytical framework capable of decomposing observed performance into separable cognitive processes.

The drift diffusion model (DDM; Ratcliff, 1978; Ratcliff & McKoon, 2008) offers such a framework. The DDM characterizes two-alternative forced-choice decisions as a process of evidence accumulation toward one of two decision boundaries. By jointly modeling the full distribution of response times and accuracy, it decomposes observed performance into four cognitively interpretable parameters—drift rate (v), reflecting the rate of information accumulation; decision threshold (a), reflecting response caution;

non-decision time (t), reflecting sensory encoding and motor execution speed; and starting point bias (z), reflecting a priori response bias (Frischkorn & Schubert, 2018; Ratcliff et al., 2016). The hierarchical Bayesian estimation framework (HDDM; Wiecki et al., 2013) further improves the stability of individual parameter estimates through group-level prior constraints. From a generative model perspective, embedding data-generating mechanisms into statistical models can help mitigate the reliability paradox inherent in difference scores (Haines et al., 2025; Hedge et al., 2018). Importantly, these four parameters offer a direct means of testing the two hypotheses—if they follow different organizational patterns at the cross-task level, low cross-task correlations are more parsimoniously explained by measurement conflation than by construct inconsistency.

Recent research applying DDM to individual differences in executive function provides key precedent. Weigard et al. (2021) extracted a task-general evidence accumulation efficiency factor from drift rates across multiple cognitive control tasks and found that it predicted self-reported self-control better than traditional difference scores. Löffler et al. (2024) further showed that when drift rates replace traditional indices in testing the Miyake three-factor structure, only a single common factor fully explained by basic information processing speed emerges. These findings suggest that the difficulty in isolating inhibition-specific factors within the Miyake framework may arise not because inhibitory processes do not exist, but because subtraction operations conflate domain-general processing efficiency with inhibition-specific mechanisms, leaving insufficient unique variance for inhibition to manifest. However, these studies combined different types of inhibitory control tasks without directly examining whether DDM parameters exhibit organizational patterns that differ between interference control and response inhibition—a question that requires simultaneously estimating all four parameters across multiple tasks of both types. Although meta-analytic evidence indicates high cross-task stability for drift rate and non-decision time (von Krause et al., 2022), and clinical studies have identified differentiated deficit patterns across parameters (Feldman & Huang-Pollock, 2021; Huang-Pollock et al., 2020; Shen et al., 2024; T. Zhang et al., 2025), these converging but disparate findings have not yet been integrated into a unified framework that systematically characterizes how different cognitive processes are organized across the two task types. The DDM parameter framework allows us to derive specific predictions that address this gap.

If the cognitive processes indexed by different parameters differ in their degree of task dependence, they should exhibit distinct cross-task organizational patterns. Drift rate, decision threshold, and non-decision time reflect general information processing characteristics—rate of evidence accumulation, response caution, and sensorimotor speed—rather than processes specific to inhibitory control. Given their demonstrated cross-task stability (Frischkorn & Schubert, 2018), we predicted that these three parameters would be shared across both task types, with no systematic differences between within-type and between-type correlation magnitudes. The critical prediction distinguishing the two task types is concerned with starting point bias (z). In interference control tasks, participants have no systematic prior preference for either response option, so *z* should remain close to the decision midpoint. Response inhibition tasks differ fundamentally—although the *Go/No-Go*, *Stop-Signal*, and *Antisaccade* tasks vary in surface features, they share a defining structural characteristic: the systematic induction of a prepotent response tendency, whether through high-frequency Go trials, competition between Go and Stop processes, or reflexive saccades triggered by peripheral cues (Munoz & Everling, 2004; Verbruggen & Logan, 2009). The DDM theory predicts that this tendency will manifest as *z* deviating from the decision midpoint (Gomez et al., 2007; Ratcliff & McKoon, 2008), and previous studies have confirmed this in individual response inhibition tasks (Weigard et al., 2020; J. Zhang et al., 2016). Mean-level shifts within single tasks, however, are insufficient to establish a stable individual differences dimension. The present study extends this logic by predicting that individuals differ stably in their ability to resist prepotent response tendencies. These individual differences should give rise to systematic covariation across tasks employing different induction mechanisms. If confirmed, this pattern would indicate that among the four DDM parameters, only *z* follows the theoretical classification of interference control versus response inhibition, while the other three transcend it—suggesting that the core computational distinction between the two task types lies not in processing efficiency or strategy, but in whether a prior response tendency must be overcome. This would provide new process-level evidence for the theoretical distinction between interference control and response inhibition—evidence that traditional indices, by conflating prior tendency modulation with general efficiency, cannot provide.

Beyond these baseline-level predictions, a further question concerns whether conflict-specific processes are shared across tasks. Traditional interference effect subtraction operations struggle to cleanly separate baseline processing efficiency from conflict-specific processes, whereas the DDM offers a conceptually cleaner quantification through the condition-difference drift rate (Δv), defined as the magnitude of drift rate reduction caused by conflict. If interference control relies on a domain-general conflict resolution mechanism, Δv should be broadly shared across tasks. If, however, different tasks engage partially different attentional selection mechanisms—semantic inhibition in *Stroop*, spatial filtering in *Flanker*, and spatial-response decoupling in *Simon*—cross-task correlations of Δv will be comparatively limited. The measurement precision of condition-difference parameters may also be constrained by the signal-to-noise ratio loss inherent in subtraction operations.

In summary, the present study used hierarchical Bayesian drift diffusion modeling to decompose the computational processes underlying inhibitory control and to test whether low cross-task correlations among traditional indices reflect construct inconsistency or the measurement conflation of variance components with different sharing properties. We employed three interference control tasks (*Stroop*, *Flanker*, *Simon*) and three response inhibition tasks (*Go/No-Go*, *Stop-Signal*, *Antisaccade*), recruiting 125 healthy adults across two independent sites. If DDM decomposition reveals that different parameters follow distinct organizational patterns—some shared broadly across task types, others selectively within one type—this would favor the measurement masking hypothesis; uniformly low cross-task sharing across all parameters would instead support the construct failure account. Building on core structural analyses, we used partial correlation analysis to reveal the process-level sources underlying low correlations among traditional indices and examined the predictive validity of computational parameters relative to traditional indices for fluid intelligence and personality traits. Cross-site replication provides an independent assessment of the generalizability of the core findings.

Methods

Participants

A total of 133 healthy adults were recruited across two independent sites in China. All participants were right-handed, had normal or corrected-to-normal vision, were free of color blindness or color weakness, and had no history of neurological or psychiatric disorders. Sample size was informed by previous studies employing similar methods (Lerche et al., 2020; Weigard et al., 2021), with approximately 65 participants at each site to support cross-site validation. All participants provided written informed consent and received monetary compensation. The study was approved by the ethics committees of Northwest Normal University and Anyang Normal University.

Measures

Demographic and Individual Difference Measures

Demographic information (e.g., age, sex, education level) and family socioeconomic status (Zhao et al., 2022; see Supplementary Methods) were collected via questionnaire. Fluid intelligence was assessed using Raven's Advanced Progressive Matrices (RAPM; Raven, 2000), a 60-item figural reasoning test in which the total number of correct responses served as the performance index. Personality traits were assessed using the Chinese Big Five Personality Inventory-Brief Version (CBF-PI-B; Meng-Cheng et al., 2011), which covers neuroticism, extraversion, openness, agreeableness, and conscientiousness (see Supplementary Methods).

Interference Control and Response Inhibition Tasks

Tasks were selected to represent classical paradigms within each inhibitory control domain while maintaining two-alternative forced-choice response formats where applicable.

Three classic interference control tasks were employed (see Figure 1A): the *Stroop* color-word task (Stroop, 1935), arrow-version *Flanker* task (Eriksen & Eriksen, 1974), and *Simon* task (Simon & Rudell, 1967). Each task included congruent and incongruent conditions in which participants made two-alternative forced-choice key-press responses to target stimuli while ignoring irrelevant information. The traditional index was the interference effect, calculated as the difference between incongruent and congruent mean response times.

Three classic response inhibition tasks were also employed (see Figure 1A): the *Go/No-Go* task (Gomez et al., 2007), *Stop-Signal* task (Logan & Cowan, 1984), and *Antisaccade* task (Munoz & Everling, 2004). In the *Go/No-Go* task, participants responded to Go stimuli while withholding responses to No-Go stimuli, with the false alarm rate serving as the traditional index. In the *Stop-Signal* task, participants inhibited an already-initiated response upon hearing a stop signal, with stop-signal reaction time (SSRT) calculated via the integration method (Verbruggen et al., 2019) serving as the traditional index. In the *Antisaccade* task, participants inhibited reflexive saccades toward peripheral cues, with the antisaccade error rate serving as the traditional index. Calculation formulas for all behavioral indices are summarized in Table S1. Detailed procedural parameters for all six tasks are provided in the Supplementary Materials.

Procedure

Participants completed all tasks in a computer laboratory equipped with individual cubicles containing 15.6-inch LED monitors (1600 × 900 pixels; 60 Hz) at a viewing distance of 60 cm. All tasks were programmed in *E*-Prime 2.0 (Psychology Software Tools, Pittsburgh, PA), and trained psychology graduate students administered the sessions. Task order was counterbalanced across participants using a Latin square design. The RAPM and Big Five questionnaire were administered after the six cognitive tasks. Participants took self-paced breaks between tasks, and the entire session lasted approximately 90 minutes.

Data Analysis

Data Preprocessing

Following standard practices (Ratcliff, 1993; Whelan, 2008), we first excluded error trials and then removed anticipatory responses (RT < 150 ms) and outliers (>3 *SD* from each participant's mean; Ambrosi et al., 2020). For the *Antisaccade* task, saccade onset was identified using a velocity threshold (>30°/s; Fischer & Weber, 1992), and trials with latencies below 80 ms or above 800 ms were excluded.

Psychometric Evaluation of Traditional Behavioral Indices

We compared demographic variables and behavioral indices between sites and computed correlations both among behavioral indices and between behavioral indices and demographic variables, with all comparisons corrected using the FDR-BH method. Split-half reliability (odd–even, Spearman–Brown corrected) was calculated for each index, and intraclass correlation coefficients (ICC) between sites assessed cross-site consistency. Confirmatory factor analysis (CFA) examined the latent structure of traditional indices by fitting both a single-factor model (all indices loading on one inhibitory control factor) and two-factor correlated model (interference control and response inhibition as separate factors). Acceptable model fit was defined as $\chi^2/df < 3$, CFI > .90, TLI > .90, RMSEA < .08, and SRMR < .08 (Hu & Bentler, 1999).

Hierarchical Drift Diffusion Modeling

Hierarchical drift diffusion modeling (HDDM; Wiecki et al., 2013) was applied to trial-level data from each task. HDDM estimates group- and individual-level parameters simultaneously within a Bayesian framework, with the hierarchical structure constraining individual estimates toward group tendencies to maintain stability while preserving meaningful individual differences. Four core parameters were estimated: drift rate (v), reflecting the rate of evidence accumulation; decision threshold (a), reflecting response caution; non-decision time (t), reflecting encoding and motor execution time; and starting point bias (z), reflecting a priori response bias. Trial-to-trial variability parameters (e.g., sv, sz, st) were included to capture intrinsic processing fluctuations (Ratcliff & McKoon, 2008).

Modeling strategies were tailored to the characteristics of each task. For interference control tasks, drift rate and decision threshold were allowed to vary freely across conditions. For the *Go/No-Go* and *Stop-Signal* tasks, a stimulus-coding scheme was employed to cast the judgment as a binary decision (Gomez et al., 2007). For the *Antisaccade* task, only correct antisaccade trials were modeled. The starting point bias was freely estimated in all tasks. Detailed coding rules and model specifications are provided in the Supplementary Methods.

Posterior distributions were sampled via Markov chain Monte Carlo (MCMC), using three independent chains of 15,000 iterations each, with the first 5,000 discarded as burn-in. Convergence was confirmed by the Gelman–Rubin statistic ($\hat{R} < 1.01$) and effective sample size (ESS > 400; Gelman & Rubin, 1992); full diagnostic criteria are provided in the Supplementary Methods. Parameter reliability was assessed using split-half correlations (Spearman–Brown corrected) and cross-site ICC. Bayesian ANOVA tested systematic parameter differences between task types (interference control vs. response inhibition), with the results reported as mean differences, 95% highest density intervals (HDI), posterior probabilities, and Bayes factors (BF_{10}) (Wagenmakers et al., 2018). As a confound check, we computed individual-level Pearson correlations between drift rate and decision threshold for each task to evaluate whether speed–accuracy tradeoff (SAT) strategies might confound subsequent cross-task sharing analyses.

Cross-Task Structural Analysis of DDM Parameters

Cross-task sharing patterns were examined at two levels: baseline processing efficiency and conflict resolution processes. For baseline condition parameters, we computed cross-task Pearson correlation matrices for the four DDM parameters under congruent/baseline conditions and compared these with correlations of traditional indices via Fisher's z transformation; reliability-corrected coefficients were also reported. Correlations were grouped by task type pairing—within-type (among interference control tasks or among response inhibition tasks) versus between-type—and differences were tested using Steiger's z and bootstrap confidence intervals. Significantly higher within-type than between-type correlations would indicate type-dependent sharing, whereas comparable magnitudes would indicate broad, type-unconstrained sharing. CFA was conducted separately for each parameter, comparing single-factor and two-factor correlated models using χ^2 difference tests (lavaan package;

Rosseel, 2012). ICC (2,1) was used to decompose total variance into cross-task shared and task-specific components.

For condition-difference parameters, we examined cross-task correlations of DDM parameters under incongruent conditions and of condition-difference (Δ) parameters to test whether conflict resolution processes per se are shared across tasks. Δv was defined as $v_{\text{congruent}} - v_{\text{incongruent}}$ (larger values indicating greater conflict-induced slowing), and Δa was defined as $a_{\text{incongruent}} - a_{\text{congruent}}$ (larger values indicating greater threshold elevation). The subtraction directions were reversed because conflict reduces drift rate but may elevate the threshold. Cross-task correlations of Δv and Δa among the three interference control tasks were computed and compared directly with those of traditional interference effects (ΔRT).

Three supplementary analyses assessed the robustness of findings. First, hierarchical cluster analysis (Ward's method, Euclidean distance) was applied to the 24 indices (6 tasks × 4 parameters) under congruent/baseline conditions. Second, a support vector machine (SVM) classifier used all DDM parameters to predict task type, with ablation analysis identifying the most discriminative parameters (see Supplementary Methods). Third, partial correlations were computed by sequentially controlling for each DDM parameter to examine how cross-task correlations of traditional indices changed, thereby isolating the relative contributions of different variance sources.

Association Analysis Between DDM Parameters and External Criteria

External validity was assessed by examining associations between the four DDM parameters and both fluid intelligence (RAPM) and the Big Five personality dimensions. At the observed-variable level, we computed Pearson correlations between each task's congruent/baseline DDM parameters and both RAPM scores and Big Five dimensions. At the latent level, factor scores from the preceding structural analyses were entered into correlation and regression analyses. Hierarchical regression tested incremental predictive validity by entering traditional indices first, followed by DDM factor scores, with ΔR^2 indexing the incremental contribution. A further step tested whether the response inhibition factor of starting point bias retained predictive value after controlling for the drift rate common factor.

All statistical tests used $\alpha = .05$ with Benjamini–Hochberg correction (Benjamini & Hochberg, 1995). All core analyses were conducted separately at each site to assess replicability. All core analyses were conducted separately at each site to assess replicability. This study was not preregistered.

Results

Sample Characteristics and Psychometric Performance of Traditional Behavioral Indices

The final sample comprised 125 participants after excluding eight for excessive missing data or failure to follow task instructions (exclusion rate: 6.0%). Participants ranged in age from 18 to 35 years (M = 22.38, SD = 3.21) and included 67 females. The primary site contributed 64 participants and the validation site contributed 61; the two sites did not differ on demographics, fluid intelligence, or any of the Big Five dimensions (all ps > .05; Table 1).

All six tasks yielded the expected behavioral patterns. For all interference control tasks, incongruent response times were significantly longer than congruent response times (*Stroop* ΔM = 78 ms; *Flanker* ΔM = 56 ms; *Simon* ΔM = 34 ms; all ps < .001), with accuracy maintained at high levels across conditions (congruent: 96%–98%; incongruent: 91%–94%). Response inhibition performance was within expected ranges (*Go/No-Go* false alarm rate: M = .23, SD = .09; SSRT: M = 241 ms, SD = 47; *Antisaccade* error rate: M = .27, SD = .11), with no significant differences between the sites (Figure 1B, Table 1).

The reliability of traditional indices was acceptable (Table 2), with mean split-half reliability of r_{SB} = .61 for interference effects and r_{SB} = .72 for response inhibition indices. Cross-task correlations, however, were notably low (Figure 1C): the mean correlation across all 15 task pairs was r = .13, with within-interference-control correlations averaging .19, within-response-inhibition correlations averaging .22, and between-type correlations averaging only .07. Reliability correction raised within-type values modestly (interference control: .29; response inhibition: .30) but left between-type correlations low (.11).

CFA revealed poor fit for the single-factor model (CFI = .72, RMSEA = .13, SRMR = .10; Table S2, Figure 1D). The two-factor correlated model showed improved fit but still fell short of acceptable standards (CFI = .87, RMSEA = .10, SRMR = .07), with a between-factor correlation of $r = .39$ and factor loadings ranging from .36 to .72, indicating substantial task-specific variance.

Cross-Task Consistency of DDM Parameters and Their Type Specificity

Cross-Type Sharing of Drift Rate, Decision Threshold, and Non-Decision Time

All HDDM models converged satisfactorily, with all $\hat{R}$ values below 1.01 (Figure S1, Table S3), and posterior predictive checks confirmed adequate data fit (Figure S1C–D). DDM parameters generally showed higher reliability than traditional indices (Table 2): mean split-half reliability was $r_{SB} = .79$ for drift rate, .76 for decision threshold, .83 for non-decision time, and .59 for starting point bias. As a confound check, drift rate and decision threshold showed only weak positive correlations at the individual level across the six tasks (mean $r = .12$, range: .05–.21; Figure 2C), suggesting that their cross-task sharing reflects independent processes rather than a single speed–accuracy tradeoff strategy.

Bayesian comparisons revealed systematic between-type differences in each parameter (Figure 2A–B). The drift rate was higher in interference control than in response inhibition tasks (M_{diff} = 0.48; 95% HDI: 0.31–0.65; BF_{10} > 30), whereas decision threshold was higher in response inhibition tasks (M_{diff} = 0.22; 95% HDI: 0.12–0.33; BF_{10} > 50). Non-decision time showed no strong overall between-type difference (M_{diff} = 0.03; 95% HDI: −0.02–0.08; BF_{10} = 0.82), although the *Antisaccade* task was notably higher than the other five tasks (M_{diff} = 0.09; 95% HDI: 0.06–0.12; BF_{10} > 100). The most pronounced between-type difference emerged for starting point bias (M_{diff} = 0.04; 95% HDI: 0.03–0.06; BF_{10} > 100): z/a was significantly above 0.50 for all response inhibition tasks (*M* range: .52–.56; all BF_{10}s > 10) but remained close to 0.50 for all interference control tasks (*M* range: .49–.51; Figure 2B).

DDM parameters under congruent/baseline conditions showed markedly higher cross-task consistency than traditional indices (Figure 2D). Non-decision time exhibited the highest consistency (mean r = .51), followed by decision threshold (mean r = .43), drift rate (mean r = .34, significantly exceeding traditional indices, p = .006), and starting point bias (mean r = .25); ICC variance decomposition yielded the same ordering (Figure 2E): .67, .59, .51, and .37, respectively.

Importantly, drift rate, decision threshold, and non-decision time all showed broad cross-type sharing with no significant difference between within-type and between-type correlations. For drift rate, mean correlations were .36 within interference control, .37 within response inhibition, and .31 between types (p = .52). Decision threshold and non-decision time showed the same patterns (ps > .45; Figure 2D). Hierarchical cluster analysis confirmed that drift rate, decision threshold, and non-decision time each clustered by parameter rather than by task type (Figure 2G). CFA provided consistent latent-variable evidence (Table S4, Figure 3A): single-factor models showed acceptable fit for all three parameters (drift rate: CFI = .97, RMSEA = .07; decision threshold: CFI = .96, RMSEA = .08; non-decision time: CFI = .99, RMSEA = .04), two-factor models did not improve fit (all $\Delta\chi^2$ ps > .10), and between-factor correlations were uniformly high (r = .78–.86).

Type-Dependent Structure of Starting Point Bias

In contrast to the other three parameters, starting point bias exhibited a distinctly type-dependent structure. Within-response-inhibition correlations (mean r = .40) were significantly higher than both between-type correlations (mean r = .20; Steiger's z = 2.54, p = .011) and within-interference-control correlations (mean r = .25; Figure 2D). ICC for starting point bias within response inhibition (.53) likewise exceeded that within interference control (.34) and between types (.23; Figure 2E); this difference held after reliability correction (p < .001) and after controlling for fluid intelligence. Hierarchical clustering revealed the same pattern (Figure 2G): unlike drift rate, decision threshold, and non-decision time, which clustered by parameter, starting point bias split by task type—response inhibition values clustered together while interference control values formed a separate cluster.

CFA confirmed this type-dependent structure at the latent level (Figure 3A). The single-factor model fit poorly (CFI = .83, RMSEA = .14), whereas the two-factor model fit significantly better ($\Delta\chi^2$ (1) = 16.86, p < .001; CFI = .96, RMSEA = .07), with a between-factor correlation of only r = .35. Response inhibition factor loadings (.51–.67) were generally higher than interference control loadings (.32–.46), indicating that starting point bias captured more systematic individual differences in response inhibition tasks. This factor showed partial overlap with the drift rate common factor (r = −.19 vs. partial r = −.11; Figure 2F), but its type-specific structure—including the two-factor model advantage and selective within-response-inhibition covariation—remained unchanged after controlling for drift rate.

Process-Level Sources of Low Cross-Task Correlations in Traditional Indices

Having established the multi-level structure of DDM parameters, next, we examined what drives the low correlations observed for traditional indices. Partial correlation analyses probed the association between low cross-task correlations of traditional indices and specific cognitive processes (Figure 3B). Congruent-condition drift rate correlated negatively with traditional interference effects (r = −.34), indicating that higher baseline efficiency was associated with smaller interference effects. Drift rates from congruent and incongruent conditions were highly correlated (r = .76), suggesting that subtraction did not adequately remove shared variance in baseline efficiency.

The critical finding concerned how cross-task correlations shifted after controlling for different parameters. For interference control tasks, correlations dropped from .19 to .08 after controlling for drift rate (p = .028), whereas controlling for decision threshold (Δr = .02), non-decision time (Δr = .01), or starting point bias (Δr = .03) produced negligible changes. For response inhibition tasks, correlations dropped from .22 to .14 after controlling for drift rate and further to .08 after additionally controlling for starting point bias. Cumulative control analyses yielded consistent results (Figure 3B), indicating that drift rate accounts for a substantial portion of the residual cross-task covariance in traditional indices.

Whereas partial correlations identified which parameters account for cross-task covariance in traditional indices, SVM ablation analysis addressed a complementary question: which parameters best distinguish the two task types? A classifier trained on all four DDM parameters across six tasks achieved an AUC of 0.79 at the primary site (Table S5). Removing the decision threshold produced the largest performance drop (ΔAUC = −0.11), followed by starting point bias (ΔAUC = −0.08), drift rate (ΔAUC = −0.05), and non-decision time (ΔAUC = −0.01; Figure 3D).

Condition-Difference Parameters and External Validity

Cross-Task Consistency of Condition-Difference Drift Rate

Cross-task patterns under incongruent conditions and for condition-difference parameters were examined next. The incongruent-condition correlation pattern largely mirrored the congruent-condition pattern (Figure 3C), although magnitudes were slightly lower: drift rate decreased from .34 to .28 (z = 1.42, p = .078), decision threshold from .43 to .39 (z = 0.93, p = .176), non-decision time from .51 to .47 (z = 1.01, p = .156), and starting point bias from .25 to .17 (z = 0.62, p = .268). All decreases were directionally consistent but none reached statistical significance, indicating that conflict conditions had limited impact on cross-task sharing patterns. Type-dependent patterns also held under incongruent conditions: drift rate, decision threshold, and non-decision time remained shared across both task types, while starting point bias continued to show selective within-response-inhibition sharing (within-type r = .36 vs. between-type r = .12, p = .004).

Condition-difference drift rate (Δv) was positive for all three interference control tasks, confirming conflict-induced slowing of evidence accumulation (Figure 3E). Cross-task correlations of Δv were low-to-moderate (mean r = .21, range: .16–.27; $r_{\text{corrected}}$ = .34), lower than congruent-condition drift rate ($r_{\text{corrected}}$ = .48; p = .042) but significantly above zero (p = .012) and slightly higher than traditional interference effects ($r_{\text{corrected}}$ = .29). Importantly, Δv and traditional interference effects did not differ significantly in cross-task consistency (p = .31), indicating comparable performance in capturing interference-specific processes.

In contrast, Δa correlations were low and nonsignificant (mean r = .08, range: .02–.14, ps > .20; $r_{\text{corrected}}$ = .13; Figure 3E), indicating that conflict-induced threshold adjustment lacked cross-task consistency. This dissociation suggests that whatever cross-task commonality exists in conflict processing is carried primarily by impairment in evidence accumulation efficiency (Δv) rather than by strategic threshold adjustment (Δa).

Associations between DDM Parameters and External Criteria

Drift rate showed the most consistent positive association with fluid intelligence across tasks (mean $r = .29$; five of six tasks reached FDR-corrected significance; Figure 3F). At the latent level, the drift rate common factor correlated $r = .40$ with RAPM ($p < .001$). Decision threshold showed weak associations with fluid intelligence (mean $r = .07$), and non-decision time showed a weak negative trend (common factor $r = -.17$, $p = .056$).

Associations with the Big Five personality dimensions were generally weak (Figure 3F). Conscientiousness correlated positively with the decision threshold common factor ($r = .21$, $p = .038$) and negatively with the response inhibition factor of starting point bias ($r = -.19$, $p = .046$). No other personality dimension reached significance with any DDM factor (all $|r|$s < .15, ps > .10).

Hierarchical regression analyses revealed that traditional indices alone did not significantly predict fluid intelligence ($R^2 = .07$, $p = .156$; Figure 3G). Adding DDM factors substantially increased predictive power ($R^2 = .22$, $\Delta R^2 = .15$, $p < .001$), with the drift rate common factor emerging as the sole significant predictor ($\beta = .33$). For conscientiousness, DDM factors added $\Delta R^2 = .06$, which fell short of significance ($p = .098$), suggesting that predictive associations with personality traits may require larger samples to detect reliably.

Cross-Site Validation

All core analyses were replicated at the validation site (n = 61; Figures S3–S4, Tables S3–S5). Traditional index cross-task correlations were similarly low (mean r = .12), and the DDM parameter rank ordering (non-decision time > decision threshold > drift rate > starting point bias) matched the primary site. Drift rate, decision threshold, and non-decision time showed broad type-unconstrained sharing, with well-fitting single-factor models (Figure S4A). Starting point bias again showed type-dependent sharing at the validation site: within-response-inhibition correlations (r = .41) exceeded between-type correlations (r = .16, p = .003), and the two-factor CFA model significantly outperformed the single-factor model (between-factor r = .37). Condition-difference drift rate correlations were low-to-moderate (mean r = .23), whereas Δa correlations were nonsignificant (mean r = .06, ps > .30; Figure S4E).

HDDM convergence was satisfactory at the validation site (Figure S2, Table S3). Partial correlations were directionally consistent with the primary site: interference effect cross-task correlations dropped by .13 after controlling for drift rate (Figure S4B). The drift rate–fluid intelligence association was significant (r = .44; Figure S4F), and hierarchical regression confirmed incremental prediction beyond traditional indices (ΔR^2 = .18, p = .002; Figure S4G). Despite some fluctuation in individual effect sizes, the consistency of effect directions and the overall pattern across sites support the replicability of core findings.

Discussion

Why do tasks designed to measure inhibitory control show such weak correlations with one another? By applying hierarchical Bayesian drift diffusion modeling to decompose six classic inhibitory control tasks, the present study illuminates the computational architecture underlying this long-standing puzzle (Figure 4). The core finding is that low cross-task correlations do not reflect a failure of the construct itself, but instead arise from a multi-level cognitive process organization that traditional measurement approaches obscure. DDM decomposition shows that different cognitive processes follow distinct cross-task patterns: drift rate, decision threshold, and non-decision time are shared across both task types; starting point bias shows relatively higher sharing within response inhibition tasks; and condition-difference drift rate exhibits only low-to-moderate correlations among interference control tasks. Traditional difference-score indices can neither preserve the cross-type shared layer of general efficiency nor isolate the type-specific layer of response tendency modulation. Instead, they compress both layers together with task-specific noise into a single value, giving rise to an appearance of construct inconsistency.

Multi-Level Process Organization of Inhibitory Control

The overarching finding is that the four DDM parameters are not uniformly shared or separated at the cross-task level; instead, they are organized into a hierarchical structure. The first level consists of domain-general processing efficiency and strategy calibration. Drift rate, decision threshold, and non-decision time all exhibited broad cross-task sharing unconstrained by the type boundary: within-type and between-type correlations did not differ significantly, confirmatory factor analyses supported single-factor models for each, and hierarchical clustering grouped indices by parameter rather than by task type. These three parameters reflect general information processing characteristics—the speed at which individuals accumulate evidence, their level of caution in setting decision criteria, and transmission efficiency of peripheral sensory-motor pathways—rather than processes specific to inhibitory control. This finding aligns with the evidence accumulation efficiency factor reported by Weigard et al. (2021), Löffler et al.'s (2024) demonstration that drift rate can absorb the common executive function factor, and meta-analytic evidence of high cross-task stability for drift rate and non-decision time (Schubert et al., 2016).

Each parameter showed distinct characteristics consistent with its theoretical interpretation. Non-decision time exhibited the highest cross-task correlations, reflecting its role as an index of peripheral processing speed (Ratcliff et al., 2016). Decision threshold also showed high consistency, supporting the view that individuals possess stable tendencies toward speed–accuracy tradeoff calibration (Lerche & Voss, 2017). Although mean thresholds were systematically higher in response inhibition than in interference control tasks—reflecting the more conservative strategy that occasional response withholding induces (Verbruggen & Logan, 2009)—individuals' relative rankings on the caution dimension remained stable across tasks. Notably, drift rate and decision threshold showed only weak positive correlations at the individual level, mitigating the concern that their cross-task sharing merely reflects different facets of a single speed–accuracy tradeoff strategy (Schubert et al., 2016; Weigard & Sripada, 2021).

Whereas these three parameters showed no sensitivity to the interference control–response inhibition distinction, starting point bias exhibited a markedly different pattern. The second level consists of type-specific response tendency modulation. Among the four DDM parameters, only starting point bias followed the theoretical classification of interference control versus response inhibition: cross-task correlations within response inhibition tasks significantly exceeded both between-type and within-interference-control correlations, the two-factor model outperformed the single-factor model, and the between-factor correlation was only .35. This pattern carries notable theoretical implications: the core computational distinction between the two task types may lie not in processing efficiency, decision strategy, or sensory-motor speed—all of which are universally shared—but in whether a prior response tendency must be systematically overcome. The reliability of starting point bias was relatively low (r_{SB} = .59), however, and its absolute cross-task correlation was also lower than that of the other three parameters; the robustness of this finding awaits further confirmation.

Beyond the empirical pattern itself, this multi-level organization sheds light on the long-standing difficulty in isolating inhibition-specific factors within the Miyake framework. Friedman et al. (2008) found that inhibition variance was completely absorbed by a common executive function factor—a finding widely interpreted as evidence against inhibitory control as a separable construct. The present results suggest an alternative possibility: the common executive function factor may primarily correspond to domain-general evidence accumulation efficiency at the first level, while inhibition-specific variance may reside at the second level—in starting point bias and its type-specific covariance structure. Subtraction operations are designed to eliminate shared variance at the first level, but they do so imprecisely: they retain residual contamination from general efficiency while failing to isolate the type-specific component at the second level. This interpretation remains speculative and requires future research to test it directly within the Miyake framework using DDM parameters in place of traditional indices.

An alternative methodological explanation also deserves consideration. The hierarchical Bayesian framework of HDDM, by shrinking individual parameters toward the group mean, may artificially inflate cross-task consistency to some degree. If shrinkage were the primary driver, however, all four parameters should exhibit similar cross-task sharing patterns rather than the differentiated pattern observed here—where drift rate, decision threshold, and non-decision time are broadly shared while starting point bias shows selective sharing only within response inhibition tasks. It is precisely this dissociation among parameters, rather than the absolute correlation level of any single parameter, that constitutes the primary basis for the core conclusions of this study.

Process-Level Sources of Low Cross-Task Correlations in Traditional Indices

Partial correlation analyses provided direct statistical evidence linking low cross-task correlations among traditional indices to specific cognitive processes. After controlling for drift rate, cross-task correlations decreased substantially in both task types (interference control: from .19 to .08; response inhibition: from .22 to .14), whereas controlling for other parameters left correlations essentially unchanged. This pattern indicates a strong statistical link between residual cross-task covariance in traditional indices and the domain-general processing efficiency reflected by drift rate.

This association reflects inherent limitations of subtraction operations. Drift rates from congruent and incongruent conditions were highly correlated across individuals ($r = .76$), a pattern consistent with the interpretation that subtraction does not adequately eliminate individual differences in general evidence accumulation efficiency. The negative correlation between congruent-condition drift rate and traditional interference effects ($r = -.34$) supports this interpretation: individuals with higher baseline efficiency tend to show smaller interference effects after subtraction. A substantial proportion of variance in traditional interference effects may therefore reflect residual general processing efficiency rather than conflict-specific processing. This inference aligns with long-standing methodological critiques of difference scores (Cronbach & Furby, 1970; Miller & Ulrich, 2013) and resonates with recent arguments that "control-specific" components in traditional executive function indices may be confounded with task-general information accumulation speed (Draheim et al., 2019; Weigard & Sripada, 2021).

The cross-task patterns of condition-difference parameters provide further insight into the specificity of conflict processing. Condition-difference drift rate (Δv) showed only low-to-moderate correlations among the three interference control tasks (mean r = .21; reliability-corrected: .34) and did not differ significantly from traditional interference effects (p = .31). This suggests that the primary advantage of DDM lies in effectively partitioning baseline efficiency rather than in providing a superior characterization of conflict-specific processes. Given that the three tasks engage partially different attentional selection mechanisms—semantic inhibition in *Stroop*, spatial filtering in *Flanker*, and spatial-response decoupling in *Simon*—the low-to-moderate correlations of Δv may also partly reflect genuine task-specific components of conflict processing (Egner, 2008; Kornblum et al., 1990). In contrast, cross-task correlations of Δa were near zero, indicating that conflict-induced threshold adjustment is highly task-specific. This dissociation suggests that whatever cross-task commonality exists in conflict processing is carried primarily by impairment in evidence accumulation efficiency rather than by strategic threshold adjustment—a distinction that traditional interference effects cannot capture.

Starting Point Bias: Preliminary Evidence for Shared Processes in Response Inhibition

Having established the structural dissociation, we now consider its mechanistic interpretation. The relatively higher cross-task sharing of starting point bias within response inhibition tasks is a theoretically informative finding that bridges previously disconnected observations. Prior research has separately reported that starting point bias deviates from the decision midpoint in response inhibition tasks (Weigard et al., 2020; J. Zhang et al., 2016) and approaches the midpoint in interference control tasks such as *Stroop* (Mulder et al., 2012; White & Poldrack, 2014), but these observations had not been integrated within a unified framework. The present study demonstrates that individual differences in this parameter not only vary by task type at the mean level but also exhibit type-selectivity at the level of covariance structure—a finding with considerably greater theoretical significance than mean-level differences alone.

Why do individuals' starting point biases covary systematically across three response inhibition tasks with substantially different surface features—key-press inhibition (*Go/No-Go*), response cancellation (*Stop-Signal*), and oculomotor inhibition (*Antisaccade*)? As predicted in the Introduction, this covariation likely reflects stable individual differences in the ability to resist prepotent response tendencies. Although different tasks induce prepotent responses through different mechanisms, individuals' capacity to overcome such tendencies may be shaped by a common underlying trait (Logan et al., 2014). Importantly, the type-specific structure of starting point bias remained unchanged after controlling for drift rate, indicating that this individual difference dimension cannot be fully explained by general cognitive efficiency. This finding nonetheless warrants cautious interpretation. The reliability of starting point bias was relatively low (r_{SB} = .59), its absolute cross-task correlation was limited (mean r = .40 within response inhibition), and the present study included only three response inhibition tasks. This finding should therefore be regarded as preliminary evidence whose robustness awaits confirmation in future studies employing more trials and a broader array of response inhibition paradigms.

Theoretical and Practical Implications

The multi-level process organization provides a unified account that integrates seemingly contradictory findings. The "disappearance" of the inhibition factor within the Miyake framework, the low cross-task correlations of traditional indices, and the "reemergence" of a response inhibition factor in large-sample studies can all be coherently explained (Shields & Yonelinas, 2025): as with any difference-score approach, incomplete elimination of general efficiency by subtraction operations leads to residual variance confounding, while capture of type-specific response tendency modulation is constrained by task-specific noise. At the measurement level, the drift rate common factor demonstrated significant incremental prediction of fluid intelligence beyond traditional behavioral indices (ΔR^2 = .15), indicating that computational modeling can extract cognitive efficiency information that traditional indices fail to capture. Fluid intelligence, however, is not a core criterion for inhibitory control; whether DDM parameters can better predict self-reported impulsivity, everyday self-control behaviors, or functional impairment in clinical populations remains an open question. Turning to clinical applications, this framework may carry implications for clinical assessment, where inhibitory control deficits are considered a transdiagnostic feature. Previous research has observed differentiated parameter-level deficit patterns in ADHD (Huang-Pollock et al., 2020). Whether the multi-level organization identified here in healthy adults—domain-general efficiency versus type-specific response tendency modulation—maps onto distinct clinical deficit profiles is an empirical question that warrants direct investigation in clinical samples.

Limitations and Future Directions

Several limitations should be acknowledged. First, DDM modeling strategies were not entirely consistent across tasks: interference control tasks aligned well with standard DDM assumptions, whereas the *Go/No-Go* task employed a stimulus-coding scheme, the *Stop-Signal* task modeled only Go trials, and the *Antisaccade* task modeled only correct antisaccade trials. These differences may affect the cross-task comparability of parameter estimates. Second, the standard DDM does not incorporate extended parameters that may be relevant to interference control (Hübner et al., 2010; White et al., 2011), such as time-varying drift rate; whether the low-to-moderate correlations of Δv are partly attributable to this limitation is worth exploring. Third, only split-half reliability was assessed; test–retest reliability of DDM parameters awaits examination through longitudinal designs. Fourth, external validity evidence came primarily from prediction of fluid intelligence; incorporating impulsivity measures or clinical sample data would provide more direct evidence for the framework's practical utility. Finally, whether the multi-level organization of DDM parameters generalizes to broader self-control domains—such as delay of gratification and emotion regulation—remains an open question.

Conclusions

By applying hierarchical Bayesian drift diffusion modeling to six classic inhibitory control tasks, this study reveals a hierarchically organized process structure underlying their notoriously low cross-task correlations: domain-general parameters (*drift rate*, *decision threshold*, *non-decision time*) are broadly shared across interference control and response inhibition tasks, whereas *starting point bias* shows selective sharing within response inhibition. Traditional difference scores compress these hierarchically distinct components into single values, obscuring shared signals. These findings favor the measurement masking hypothesis over the construct fragmentation hypothesis: cross-task shared processes do exist but become visible only when computational methods separate distinct cognitive components. More broadly, these results suggest that the field's difficulty in establishing inhibitory control as a coherent construct may stem less from the construct itself than from the tools used to measure it.

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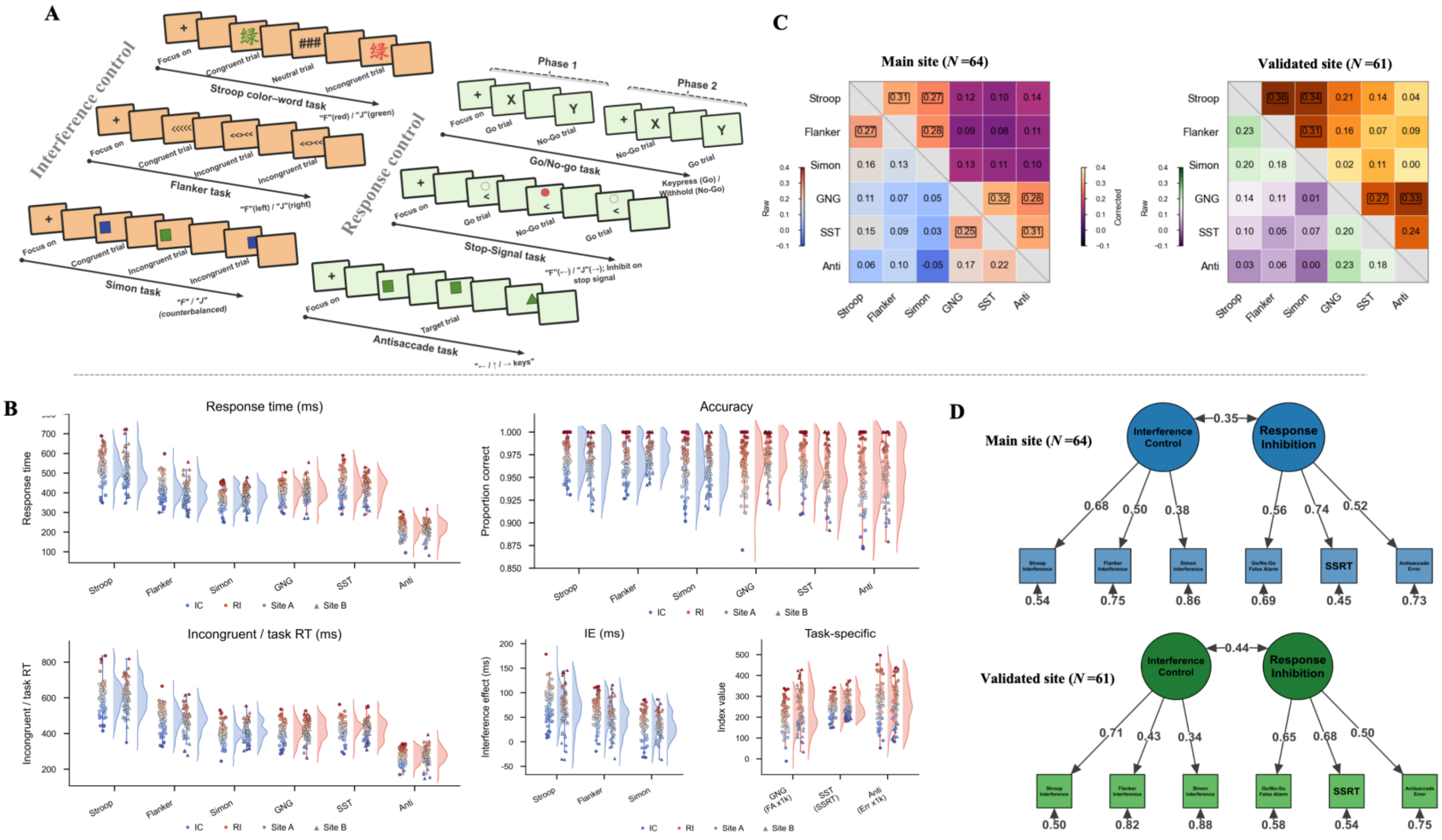
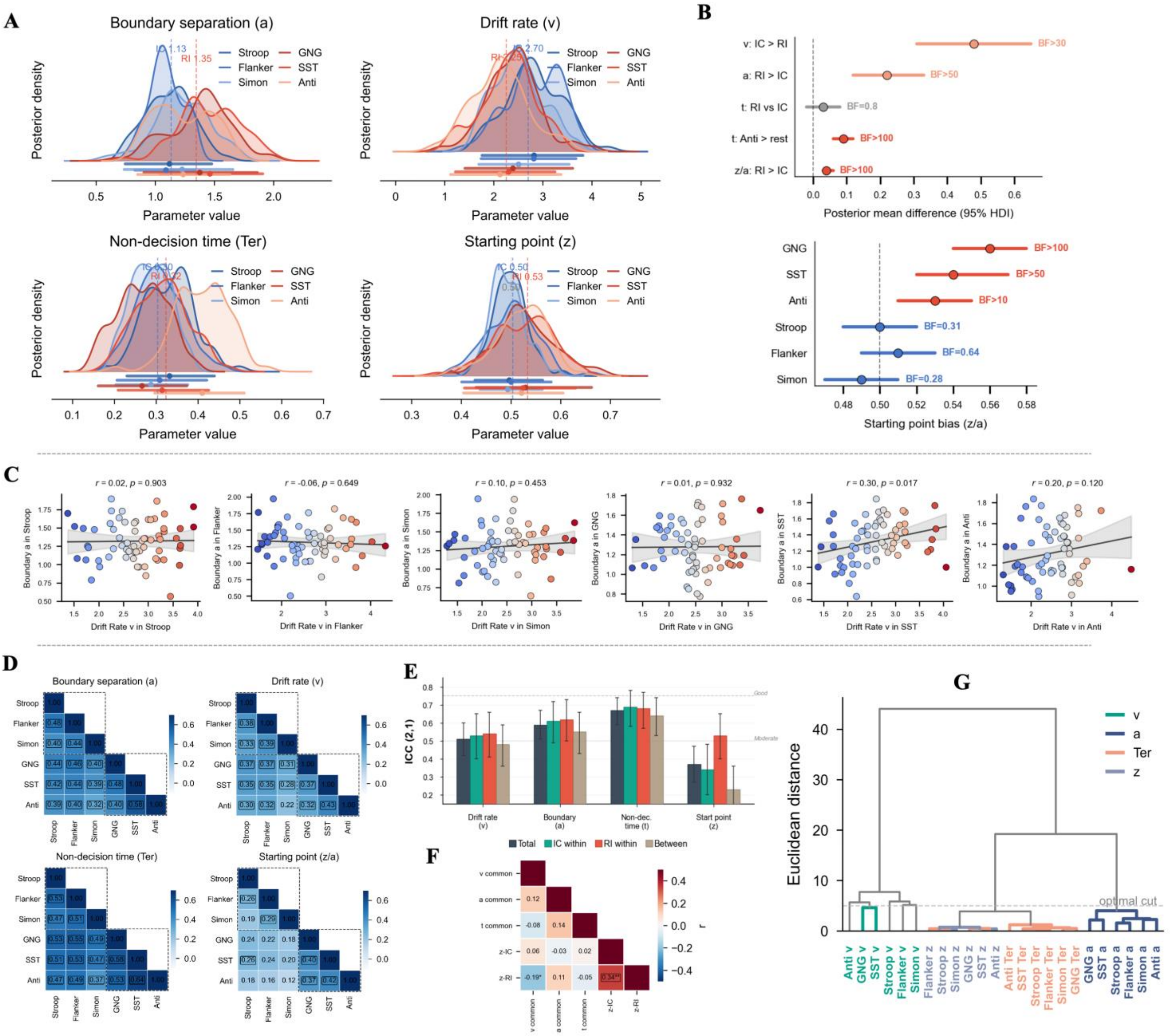
Figure 1. Experimental paradigms and psychometric properties of inhibitory control indices



Note: (A) Schematic illustration of trial sequences for the six inhibitory control tasks. Orange shading denotes interference control tasks (*Stroop*, *Flanker*, *Simon*); green shading denotes response inhibition tasks (*Go/No-Go*, *Stop-Signal*, *Antisaccade*). ITI = inter-trial interval; SSD = stop-signal delay. (B) Raincloud plots of traditional behavioral indices for each task. Individual data points represent participants; density curves depict sample distributions; embedded box plots show medians and interquartile ranges. Circles indicate primary site participants; triangles indicate validation site participants. Color gradient reflects value magnitude (red = high; blue = low). (C) Cross-task correlation matrices of traditional behavioral indices (left: primary site; right: validation site). Lower triangles display raw Pearson correlations (green = positive, purple = negative; darker shades indicate stronger correlations). Upper triangles display reliability-corrected coefficients (brown = positive; light yellow = weak). (D) Confirmatory factor analysis path diagrams for traditional indices (two-factor correlated model). Ellipses represent latent factors; rectangles represent observed variables. Single-headed arrows display standardized factor loadings; double-headed arrows display between-factor correlations. Model fit indices appear below each panel. Blue = primary site; green = validation site. IC = interference control; RI = response inhibition; Str = *Stroop*; Fla = *Flanker*; Sim = *Simon*; GNG = *Go/No-Go*; SST = *Stop-Signal* task; Anti = *Antisaccade*.

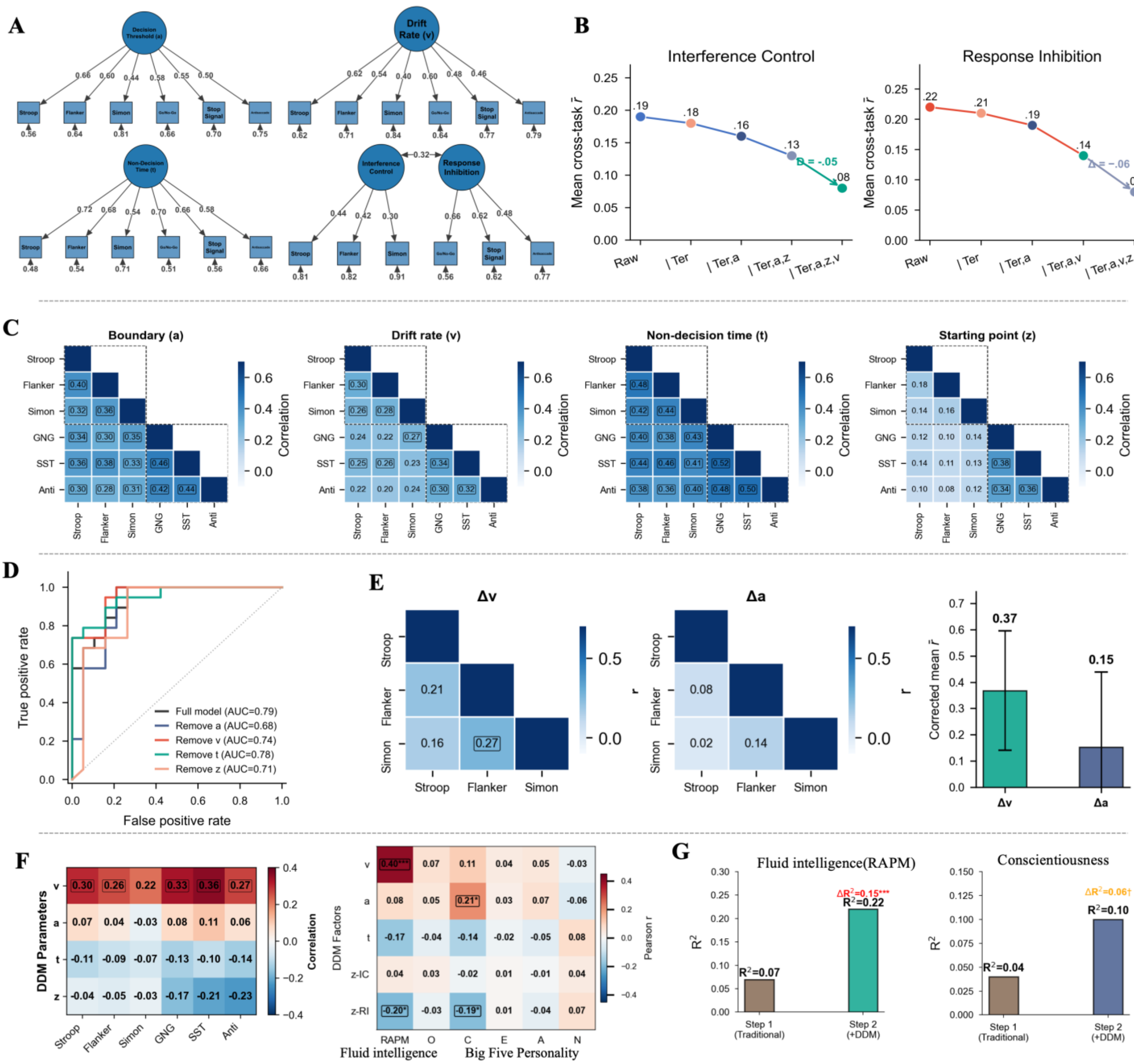
Figure 2. Cross-task sharing patterns and type specificity of DDM parameters



Note: (A) Group-level posterior distributions of the four DDM parameters across six tasks. Blue = interference control tasks; orange = response inhibition tasks. (B) Upper panel: Posterior distributions of mean parameter differences between task types. Solid vertical lines indicate posterior means; shaded regions indicate 95% highest density intervals (HDI); dashed vertical lines mark zero. Bayes factors (BF_{10}) appear in the upper right corner of each panel. Lower panel:

Posterior distributions of starting point bias (z/a) deviation from the unbiased midpoint (0.50) for interference control tasks (top row) and response inhibition tasks (bottom row). Red dashed lines indicate z/a = 0.50. (C) Scatter plots of drift rate versus decision threshold for each task, with regression lines and 95% confidence bands. (D) Cross-task correlation matrices for each DDM parameter across six tasks. Blue dashed boxes highlight within-interference-control correlations; orange dashed boxes highlight within-response-inhibition correlations. (E) Variance decomposition based on intraclass correlation coefficients (ICC). Four ICC types are shown for each parameter: overall, within-interference-control, within-response-inhibition, and between-type. Error bars represent bootstrap 95% confidence intervals. (F) Correlation heatmap among DDM latent factors, including drift rate, decision threshold, and non-decision time common factors, and interference control and response inhibition factors of starting point bias. (G) Hierarchical clustering dendrogram of 24 variables (six tasks × four parameters) using Ward's method with Euclidean distance. Leaf nodes are colored by parameter type; dashed boxes highlight the type-specific clustering of starting point bias. v = drift rate; a = decision threshold; t = non-decision time; z = starting point bias; IC = interference control; RI = response inhibition. n.s. = not significant; *p < .05; **p < .01.

Figure 3. Latent structure, process-level associations, and external validity of

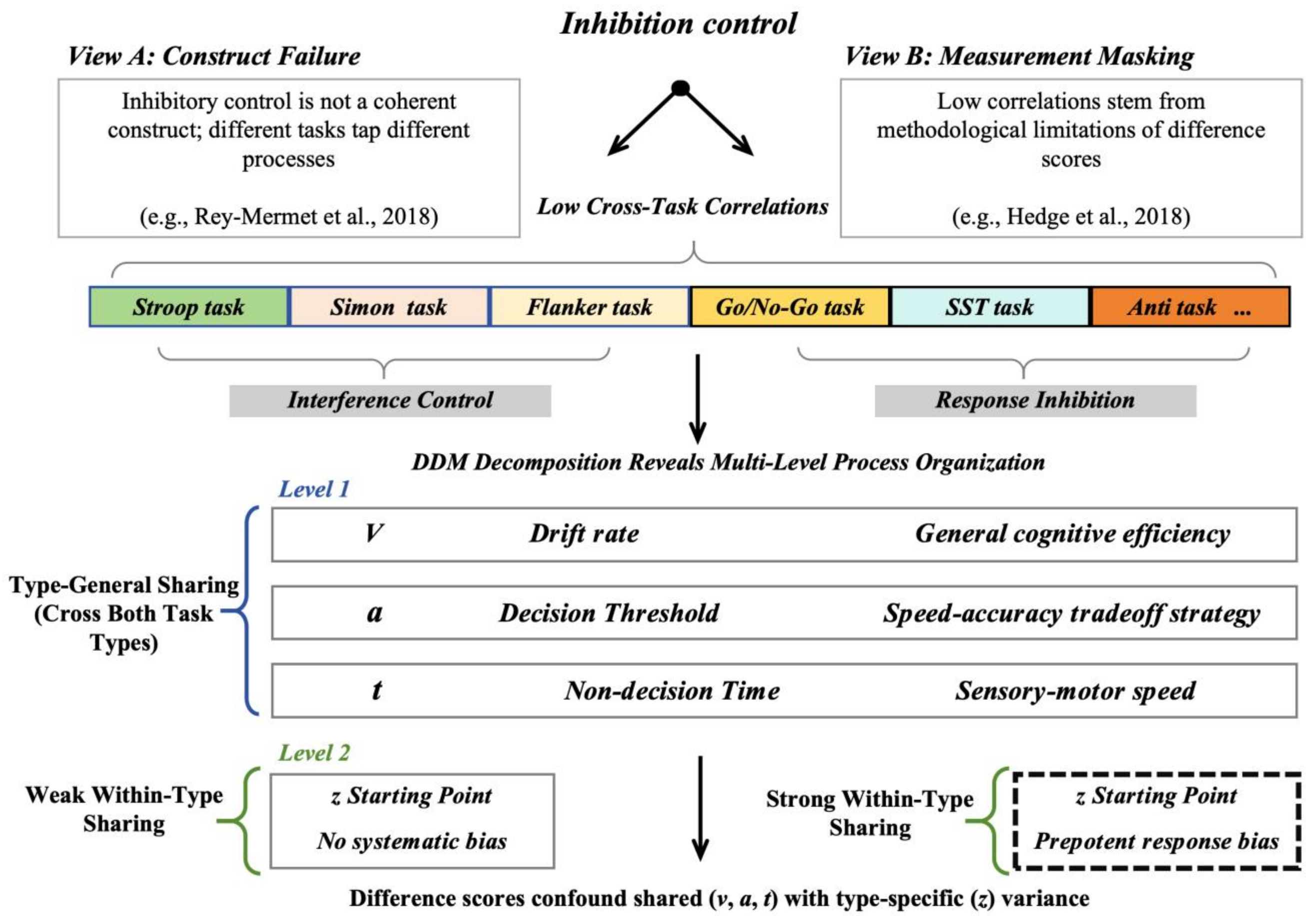


DDM parameters

Note: (A) Confirmatory factor analysis path diagrams for each DDM parameter. Ellipses represent latent factors; rectangles represent observed variables. Single-headed arrows display standardized factor loadings; double-headed arrows display between-factor correlations. Model fit indices appear below each panel. (B) Partial correlation analysis showing changes in cross-task correlations among traditional indices after sequentially controlling for DDM parameters. Left panel: interference control tasks; right panel: response inhibition tasks. X-axis labels indicate cumulative controls: Raw (no control), then sequential addition of non-decision time ($|t$), decision threshold ($|a$), starting point bias ($|z$), and drift rate ($|v$). Error bars represent standard errors. (C) Cross-task correlation matrices for DDM parameters estimated from incongruent/task conditions. Dashed boxes delineate interference control (IC) and response inhibition (RI) task clusters. Boxed values indicate significant correlations ($p < .05$). (D) ROC curves from SVM classification with ablation analysis. Black curve = full model; colored curves = performance after removing each parameter. (E) Cross-task correlation matrices for condition-difference parameters (Δv, Δa) among interference control tasks. Diagonal values are Spearman–Brown corrected split-half reliabilities. Bar charts compare mean reliability-corrected cross-task correlations for Δv and Δa; error bars represent bootstrap 95% confidence intervals. (F) Left panel: Correlations between DDM latent factors and fluid intelligence (RAPM) with 95% confidence intervals. Right panel: Correlation matrix of DDM factors with external criteria (RAPM and Big Five personality dimensions). Rows represent DDM factor scores (v, a, t as common factors; z separated into IC and RI factors); columns represent external criteria. Color intensity and values indicate Pearson correlations; asterisks indicate FDR-corrected significance. (G) Hierarchical regression predicting fluid intelligence (left) and conscientiousness (right). Step 1 includes traditional indices only; Step 2 adds DDM latent factors. Bar height represents R^2; ΔR^2 and significance levels are annotated. Str = *Stroop*; Fla = *Flanker*; Sim = *Simon*; GNG = *Go/No-Go*; SST = *Stop-Signal* task; Anti = *Antisaccade*; RAPM = Raven's Advanced

Progressive Matrices; N = Neuroticism; E = Extraversion; O = Openness; A = Agreeableness; C = Conscientiousness. †$p < .10$; *$p < .05$; **$p < .01$; ***$p < .001$.

Figure 4. A DDM-based framework for understanding low cross-task correlations in inhibitory control



Note: The upper portion illustrates two prevailing theoretical accounts of low cross-task correlations: the construct failure hypothesis (View A), which holds that different tasks engage qualitatively distinct processes, and the measurement masking hypothesis (View B), which attributes low correlations to the methodological limitations of difference scores. The middle portion depicts the six tasks organized by inhibitory control subtype. The lower portion presents the core findings of the present study within a multi-level process framework. At Level 1, drift rate (v), decision threshold (a), and non-decision time (t) exhibit broad, type-general sharing across both interference control and response inhibition tasks. At Level 2, starting point bias (z) shows selective sharing only within response inhibition tasks—where task structures systematically induce a prepotent response tendency—but not within interference control tasks. Traditional difference scores conflate these shared (v, a, t) and type-specific (z) variance components into a single value, obscuring the multi-level organization that underlies cross-task consistency. SST = Stop-Signal task; Anti = Antisaccade task.

Table 1. Sample characteristics and behavioral performance: Descriptive statistics and cross-site comparisons

Variable	Primary Site (n = 64)	Validation Site (n = 61)	t/χ^2	p	Effect Size
Demographic Variables					
Age (years)	22.25 ± 3.15	22.52 ± 3.28	0.47	.640	0.08
Sex (male/female)	30/34	28/33	0.01	.913	0.01
Years of education	14.69 ± 1.81	14.85 ± 1.72	0.51	.614	0.09
Raven's fluid intelligence	51.42 ± 5.68	50.85 ± 5.94	0.55	.584	0.10
Big Five Personality					
Neuroticism	24.42 ± 6.48	25.34 ± 6.72	0.78	.437	0.14
Extraversion	29.38 ± 6.02	29.57 ± 5.94	0.18	.859	0.03
Openness	31.22 ± 5.42	30.54 ± 5.68	0.68	.495	0.12
Agreeableness	32.64 ± 4.86	31.79 ± 4.72	0.99	.324	0.18
Conscientiousness	30.08 ± 5.72	30.52 ± 5.36	0.44	.658	0.08
Interference Control Tasks					
***Stroop* Task**					
Congruent RT (ms)	518 ± 75	528 ± 82	0.71	.478	0.13
Incongruent RT (ms)	597 ± 88	605 ± 83	0.52	.602	0.09
Congruent accuracy (%)	97.2 ± 2.4	96.8 ± 2.8	0.86	.392	0.15
Incongruent accuracy (%)	92.4 ± 4.5	91.6 ± 4.0	1.05	.296	0.19
Interference effect (ms)	79 ± 37	77 ± 41	0.29	.775	0.05
***Flanker* Task**					
Congruent RT (ms)	416 ± 62	420 ± 57	0.37	.708	0.07
Incongruent RT (ms)	474 ± 68	473 ± 72	0.08	.936	0.01
Congruent accuracy (%)	97.4 ± 2.1	97.8 ± 1.9	1.11	.267	0.20
Incongruent accuracy (%)	93.6 ± 3.8	92.8 ± 4.3	1.10	.272	0.20
Interference effect (ms)	58 ± 31	53 ± 29	0.93	.354	0.17
***Simon* Task**					
Congruent RT (ms)	373 ± 49	378 ± 44	0.60	.550	0.11
Incongruent RT (ms)	410 ± 52	408 ± 55	0.21	.835	0.04
Congruent accuracy (%)	96.5 ± 2.7	96.3 ± 3.0	0.39	.696	0.07
Incongruent accuracy (%)	92.4 ± 4.4	93.0 ± 4.8	0.73	.467	0.13
Interference effect (ms)	37 ± 26	30 ± 24	1.56	.121	0.28
Response Inhibition Tasks					
***Go/No-Go* Task**					
Go trial RT (ms)	407 ± 52	415 ± 58	0.81	.418	0.15
Go trial accuracy (%)	96.8 ± 3.1	97.2 ± 2.6	0.78	.437	0.14
No-Go trial accuracy (%)	77.8 ± 8.2	76.4 ± 9.1	0.90	.368	0.16
False alarm rate	.22 ± .08	.24 ± .09	1.31	.191	0.24

***Stop-Signal* Task**					
Go trial RT (ms)	422 ± 59	418 ± 55	0.39	.696	0.07
Go trial accuracy (%)	96.2 ± 2.8	95.9 ± 3.0	0.58	.564	0.10
Stop success rate (%)	50.8 ± 4.2	50.4 ± 4.4	0.52	.604	0.09
SSD (ms)	174 ± 50	171 ± 47	0.35	.730	0.06
SSRT (ms)	239 ± 44	244 ± 49	0.60	.549	0.11
***Antisaccade* Task**					
Prosaccade latency (ms)	218 ± 39	216 ± 36	0.30	.767	0.05
Antisaccade latency (ms)	273 ± 46	282 ± 52	1.03	.307	0.18
Antisaccade error rate	.28 ± .11	.26 ± .10	1.06	.290	0.19

Note: Continuous variables are presented as mean ± standard deviation. Interference effect = incongruent RT − congruent RT. SSD = stop-signal delay; SSRT = stop-signal reaction time; false alarm rate = proportion of commission errors on No-Go trials; antisaccade error rate = proportion of direction errors on antisaccade trials. Between-site comparisons used independent-samples t tests for continuous variables and chi-square tests for sex. Effect sizes are reported as Cohen's d (continuous) or Cramér's V (categorical).

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Table 2. Reliability of traditional behavioral indices and HDDM parameter estimates

Category	Task	Measure	Primary site r_{SB}	95% CI	Validation site r_{SB}	95% CI	Total sample ICC	95% CI
Interference control	*Stroop*	Interference effect	.68	[.52, .79]	.65	[.48, .77]	.67	[.56, .76]
	Flanker	Interference effect	.60	[.42, .73]	.64	[.47, .76]	.62	[.50, .72]
	Simon	Interference effect	.55	[.36, .70]	.52	[.32, .67]	.54	[.41, .65]
			Mean	.61		.60		.61
Response inhibition	*Go/No-Go*	False alarm rate	.73	[.58, .83]	.70	[.54, .81]	.72	[.62, .80]
	Stop-Signal	SSRT	.76	[.62, .85]	.79	[.66, .87]	.77	[.68, .84]
	Antisaccade	Error rate	.66	[.49, .78]	.71	[.56, .82]	.68	[.57, .77]
			Mean	.72		.73		.72
Drift rate (v)	*Stroop*		.82	[.70, .89]	.85	[.74, .91]	.83	[.76, .89]
	Flanker		.79	[.66, .88]	.81	[.68, .89]	.80	[.72, .86]
	Simon		.74	[.59, .84]	.72	[.56, .83]	.73	[.63, .81]
	Go/No-Go		.83	[.72, .90]	.80	[.67, .88]	.82	[.74, .87]
	Stop-Signal		.78	[.64, .87]	.84	[.73, .91]	.81	[.73, .87]
	Antisaccade		.75	[.60, .85]	.78	[.64, .87]	.76	[.66, .83]
			Mean	.79		.80		.79
	Stroop		.80	[.67, .88]	.78	[.64, .87]	.79	[.70, .85]

	Flanker	.76	[.62, .85]	.80	[.67, .88]	.78	[.69, .85]
	Simon	.69	[.52, .81]	.72	[.56, .83]	.70	[.59, .79]
Decision threshold (a)	*Go/No-Go*	.77	[.63, .86]	.74	[.59, .84]	.76	[.66, .83]
	Stop-Signal	.81	[.68, .89]	.83	[.72, .90]	.82	[.74, .88]
	Antisaccade	.72	[.56, .83]	.68	[.51, .80]	.70	[.59, .79]
		Mean	.76		.76		.76
	Stroop	.86	[.76, .92]	.88	[.79, .93]	.87	[.81, .91]
	Flanker	.84	[.73, .91]	.82	[.70, .89]	.83	[.76, .88]
	Simon	.79	[.66, .88]	.81	[.68, .89]	.80	[.72, .86]
Non-decision time (t)	*Go/No-Go*	.83	[.72, .90]	.85	[.74, .91]	.84	[.77, .89]
	Stop-Signal	.87	[.78, .93]	.84	[.73, .91]	.86	[.80, .90]
	Antisaccade	.80	[.67, .88]	.77	[.63, .86]	.79	[.70, .85]
		Mean	.83		.83		.83
	Stroop	.54	[.34, .69]	.58	[.39, .72]	.56	[.43, .67]
	Flanker	.57	[.38, .71]	.52	[.32, .67]	.55	[.42, .66]
	Simon	.45	[.24, .62]	.49	[.29, .65]	.47	[.33, .59]
Starting point bias (z)	*Go/No-Go*	.68	[.52, .80]	.65	[.48, .77]	.67	[.55, .76]
	Stop-Signal	.64	[.47, .76]	.72	[.57, .83]	.67	[.56, .76]
	Antisaccade	.59	[.40, .73]	.63	[.45, .76]	.61	[.49, .71]
		Mean	.58		.60		.59

Note: r_{SB} = Spearman–Brown corrected split-half reliability (odd–even split). ICC = intraclass correlation coefficient, ICC (2,1), indexing cross-site consistency. 95% CI = 95% confidence interval. Traditional indices include interference effects for the three interference control tasks and the primary performance measure for each response inhibition task (*Go/No-Go*: false alarm rate; *Stop-Signal*: SSRT; *Antisaccade*: error rate). DDM parameters were estimated from congruent/baseline condition trials.